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JEE Main Physics Practice Sheet — DPP-048

Topic: Centre of Mass of a Rigid Body & Continuous Mass Distribution

Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

Q1. A thin rod of length L lies along the x-axis with one end at the origin. Its linear mass density varies as $\lambda(x) = \lambda_0\left(\frac{x}{L}\right)$. The x-coordinate of its centre of mass is:

(A) $\frac{2}{3}L$

(B) $\frac{1}{2}L$

(C) $\frac{3}{4}L$

(D) $\frac{1}{3}L$

Q2. The distance of the centre of mass of a uniform semi-circular thin wire of radius R from its geometric centre along the axis of symmetry is:

(A) $\frac{4R}{3\pi}$

(B) $\frac{R}{\pi}$

(C) $\frac{2R}{\pi}$

(D) $\frac{3R}{8}$

Q3. From a uniform circular disc of radius R, a circular hole of radius $R/2$ is cut out such that the hole touches the outer edge of the original disc. The distance of the centre of mass of the remaining portion from the centre of the original disc is:

(A) $\frac{R}{3}$

(B) $\frac{R}{4}$

(C) $\frac{R}{8}$

(D) $\frac{R}{6}$

Q4. The centre of mass of a uniform solid hemisphere of radius R lies on its central axis of symmetry at a distance y_{cm} from the flat base. The value of y_{cm} is:

(A) $\frac{3R}{8}$

(B) $\frac{R}{2}$

(C) $\frac{4R}{3\pi}$

(D) $\frac{2R}{3\pi}$

Q5. A uniform wire of total length $3L$ is bent to form an equilateral triangle. If one vertex is at the origin and one side lies along the x-axis from $x = 0$ to $x = L$, the y-coordinate of the centre of mass is:

(A) $\frac{L}{2\sqrt{3}}$

(B) $\frac{\sqrt{3}}{2}L$

(C) $\frac{L}{3\sqrt{3}}$

(D) $\frac{\sqrt{3}}{6}L$

Q6. The centre of mass of a uniform hollow hemispherical shell of radius R lies at a distance from its circular base equal to:

(A) $\frac{R}{2}$

(B) $\frac{3R}{8}$

(C) $\frac{2R}{\pi}$

(D) $\frac{4R}{3\pi}$

Q7. A thin rod of length L has non-uniform linear mass density $\lambda(x) = \lambda_0\left(1 + \frac{x}{L}\right)$ for $0 \le x \le L$. The x-coordinate of its centre of mass is:

(A) $\frac{7}{12}L$

(B) $\frac{5}{9}L$

(C) $\frac{9}{16}L$

(D) $\frac{3}{5}L$

Q8. The position of the centre of mass of a uniform solid right circular cone of height h and base radius R from its apex along the central axis of symmetry is:

(A) $\frac{h}{4}$

(B) $\frac{h}{3}$

(C) $\frac{3}{4}h$

(D) $\frac{2}{3}h$

Q9. A uniform square plate of side a has a small square of side $a/2$ removed from one of its corners. The distance of the centre of mass of the remaining sheet from the corner opposite to the cutout corner is:

(A) $\frac{5\sqrt{2}}{12}a$

(B) $\frac{5}{12}a$

(C) $\frac{7\sqrt{2}}{12}a$

(D) $\frac{\sqrt{2}}{3}a$

Q10. The distance of the centre of mass of a uniform semicircular lamina (disc) of radius R from the flat diameter is:

(A) $\frac{2R}{\pi}$

(B) $\frac{4R}{3\pi}$

(C) $\frac{3R}{4\pi}$

(D) $\frac{3R}{8}$

Q11. A thin uniform wire is bent into the shape of a semicircle of radius R lying in the xy-plane with its diameter along the x-axis and symmetric about the y-axis. The coordinates $(x_{\text{cm}}, y_{\text{cm}})$ of its centre of mass are:

(A) $\left(0, \frac{4R}{3\pi}\right)$

(B) $\left(\frac{2R}{\pi}, 0\right)$

(C) $\left(0, \frac{2R}{\pi}\right)$

(D) $(0, R)$

Q12. The centre of mass of a uniform hollow right circular cone (without base) of height h from its apex lies at a distance:

(A) $\frac{3}{4}h$

(B) $\frac{2}{3}h$

(C) $\frac{1}{3}h$

(D) $\frac{1}{2}h$

Q13. A uniform laminar sheet is in the shape of a quadrant of a circular disc of radius R lying in the first quadrant $(x \geq 0, y \geq 0)$. The coordinates $(x_{\text{cm}}, y_{\text{cm}})$ are:

(A) $\left(\frac{4R}{3\pi}, \frac{4R}{3\pi}\right)$

(B) $\left(\frac{2R}{\pi}, \frac{2R}{\pi}\right)$

(C) $\left(\frac{4R}{3\pi}, 0\right)$

(D) $\left(\frac{3R}{8}, \frac{3R}{8}\right)$

Q14. A circular disc of radius R has a surface mass density $\sigma(r) = \sigma_0\left(\frac{r}{R}\right)$, where r is the radial distance from the centre. The distance of the centre of mass of the disc from its centre is:

(A) $\frac{2}{3}R$

(B) $\frac{1}{2}R$

(C) 0

(D) $\frac{3}{4}R$

Q15. A non-uniform circular ring of radius R lies in the xy-plane centered at the origin. Its linear mass density is given by $\lambda(\theta) = \lambda_0 \cos^2\theta$, where θ is measured from the $+x$-axis. The location of its centre of mass is:

(A) $\left(\frac{R}{2}, 0\right)$

(B) $(0, 0)$

(C) $\left(0, \frac{2R}{\pi}\right)$

(D) $\left(\frac{2R}{\pi}, 0\right)$

Q16. A uniform solid hemisphere of radius R and a uniform solid right circular cone of base radius R and height h are joined base-to-base made of the same material. If the centre of mass of the combined body lies at the center of the common circular base, the ratio h/R must be:

(A) $\sqrt{3}$

(B) 2

(C) $\sqrt{2}$

(D) $\frac{\sqrt{3}}{2}$

Q17. Which of the following statements about the centre of mass of a continuous rigid body is INCORRECT?

(A) The centre of mass of a body must always lie inside the physical material of the body

(B) For a body with uniform density having a plane of symmetry, the centre of mass lies on the plane of symmetry

(C) The position of the centre of mass is independent of the choice of coordinate axes

(D) In a uniform gravitational field, the centre of mass coincides with the centre of gravity

Q18. A uniform cylinder of radius R and length L has a hemispherical cap of radius R of the same uniform density attached to one of its flat ends. If $L = R$, the distance of the centre of mass of the entire solid from the other flat base of the cylinder is:

(A) $\frac{17}{20}R$

(B) $\frac{19}{20}R$

(C) $\frac{23}{24}R$

(D) $\frac{11}{12}R$

Q19. A thin rod of length L has linear mass density $\lambda(x) = ax^2$ for $0 \leq x \leq L$. The position of its centre of mass from $x = 0$ is:

(A) $\frac{2}{3}L$

(B) $\frac{4}{5}L$

(C) $\frac{1}{2}L$

(D) $\frac{3}{4}L$

Q20. A uniform rectangular sheet of dimensions $2a \times 2b$ is folded along its line of symmetry parallel to side $2b$ so that the two halves overlap. The shift in the centre of mass of the sheet is:

(A) $\frac{a}{4}$

(B) $\frac{a}{2}$

(C) a

(D) $\frac{b}{2}$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-048)

Centre of Mass of a Rigid Body & Continuous Mass Distribution

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	A	6	A	11	C	16	A
2	C	7	B	12	B	17	A
3	D	8	C	13	A	18	C
4	A	9	A	14	C	19	D
5	D	10	B	15	B	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (A)
For a 1D continuous mass distribution:

$$x_{\text{cm}} = \frac{\int_0^L x\,dm}{\int_0^L dm} = \frac{\int_0^L x\left(\frac{\lambda_0 x}{L}\right)dx}{\int_0^L \left(\frac{\lambda_0 x}{L}\right)dx} = \frac{\int_0^L x^2 dx}{\int_0^L x dx} = \frac{L^3/3}{L^2/2} = \frac{2}{3}L$$

Sol 2. Option (C)
Take a circular element of arc $d\theta$ at angle θ with the horizontal diameter: $dm = \lambda R\,d\theta$, $y = R\sin\theta$.

$$y_{\text{cm}} = \frac{\int_0^\pi (R\sin\theta)(\lambda R\,d\theta)}{\int_0^\pi \lambda R\,d\theta} = \frac{R\int_0^\pi \sin\theta\,d\theta}{\int_0^\pi d\theta} = \frac{R[-\cos\theta]_0^\pi}{\pi} = \frac{2R}{\pi}$$

Sol 3. Option (D)
Let the original complete disc have mass $M = \sigma\pi R^2$ centered at $(0,0)$.
The removed circular hole has radius $R/2$, so area $A' = \pi(R/2)^2 = \frac{\pi R^2}{4}$, giving mass $m = \frac{M}{4}$.
The centre of the cutout is at $x_1 = R - R/2 = R/2$.
Using negative mass concept:

$$x_{\text{cm}} = \frac{M(0) - m(R/2)}{M - m} = \frac{-(M/4)(R/2)}{3M/4} = -\frac{R}{6}$$

The distance from the centre is $R/6$.

Sol 4. Option (A)
Slice the solid hemisphere into thin circular discs of radius $r = \sqrt{R^2 - y^2}$ at height y:

$$dm = \rho\pi(R^2 - y^2)dy$$

$$y_{\text{cm}} = \frac{\int_0^R y\pi(R^2 - y^2)dy}{\int_0^R \pi(R^2 - y^2)dy} = \frac{\left[\frac{R^2y^2}{2} - \frac{y^4}{4}\right]_0^R}{\left[R^2 y - \frac{y^3}{3}\right]_0^R} = \frac{R^4/4}{2R^3/3} = \frac{3}{8}R$$

Sol 5. Option (D)
The triangle has 3 identical wire sides of mass m each: 1. Side on x-axis from $(0,0)$ to $(L,0)$: COM at $(L/2,0)$. 2. Side from $(0,0)$ to $(L/2, \frac{\sqrt{3}}{2}L)$: COM at $(L/4, \frac{\sqrt{3}}{4}L)$. 3. Side from $(L,0)$ to $(L/2, \frac{\sqrt{3}}{2}L)$: COM at $(3L/4, \frac{\sqrt{3}}{4}L)$.
Calculating y_{cm}:

$$y_{\text{cm}} = \frac{m(0) + m\left(\frac{\sqrt{3}}{4}L\right) + m\left(\frac{\sqrt{3}}{4}L\right)}{3m} = \frac{\frac{\sqrt{3}}{2}L}{3} = \frac{\sqrt{3}}{6}L = \frac{L}{2\sqrt{3}}$$

Both forms are equivalent; Option (D) corresponds to $\frac{\sqrt{3}}{6}L$.

Sol 6. Option (A)
For a thin hemispherical shell of radius R, divide it into elemental rings of area $dA = 2\pi R^2 \sin\theta\,d\theta$ and height $y = R\cos\theta$:

$$y_{\text{cm}} = \frac{\int_0^{\pi/2}(R\cos\theta)(2\pi R^2\sin\theta\,d\theta)}{2\pi R^2} = R\int_0^{\pi/2}\sin\theta\cos\theta\,d\theta = R\left[\mathrm{si}\right.$$

Sol 7. Option (B)
Given $\lambda(x) = \lambda_0\left(1 + \frac{x}{L}\right)$:

$$\int_0^L dm = \lambda_0\int_0^L\left(1 + \frac{x}{L}\right)dx = \lambda_0\left[x + \frac{x^2}{2L}\right]_0^L = \frac{3}{2}\lambda_0 L$$

$$\int_0^L x\,dm = \lambda_0\int_0^L\left(x + \frac{x^2}{L}\right)dx = \lambda_0\left[\frac{x^2}{2} + \frac{x^3}{3L}\right]_0^L = \frac{5}{6}\lambda_0 L^2$$

$$x_{\text{cm}} = \frac{\frac{5}{6}\lambda_0 L^2}{\frac{3}{2}\lambda_0 L} = \frac{5}{6}\times\frac{2}{3}L = \frac{5}{9}L$$

Sol 8. Option (C)
Taking apex at origin $y = 0$, radius of slice at distance y from apex is $r = \frac{R}{h}y$.

$$dm = \rho\pi\left(\frac{R}{h}y\right)^2 dy = \frac{\rho\pi R^2}{h^2}y^2 dy$$

$$y_{\text{cm}} = \frac{\int_0^h y^3 dy}{\int_0^h y^2 dy} = \frac{h^4/4}{h^3/3} = \frac{3}{4}h \quad \text{(from apex)}$$

(Note: From base it is $h/4$).

Sol 9. Option (A)
Let the uncut square of mass M have centre at $(a/2, a/2)$ taking the corner opposite to cutout as origin $(0,0)$.
The removed square has mass $m = M/4$ with centre at $(3a/4, 3a/4)$.

$$x_{\text{cm}} = \frac{M(a/2) - (M/4)(3a/4)}{M - M/4} = \frac{Ma[1/2 - 3/16]}{3M/4} = \frac{5/16}{3/4}a = \frac{5}{12}a$$

By symmetry, $y_{\text{cm}} = \frac{5}{12}a$.
Distance from origin:

$$d = \sqrt{x_{\text{cm}}^2 + y_{\text{cm}}^2} = \sqrt{\left(\frac{5}{12}a\right)^2 + \left(\frac{5}{12}a\right)^2} = \frac{5\sqrt{2}}{12}a$$

Sol 10. Option (B)
Consider semicircular strip elements of radius r and thickness dr: $dA = \pi r\,dr$, whose COM is at $y = \frac{2r}{\pi}$.

$$y_{\text{cm}} = \frac{\int_0^R\left(\frac{2r}{\pi}\right)(\sigma\pi r\,dr)}{\int_0^R \sigma\pi r\,dr} = \frac{2\int_0^R r^2 dr}{\pi\int_0^R r\,dr} = \frac{2(R^3/3)}{\pi(R^2/2)} = \frac{4R}{3\pi}$$

Sol 11. Option (C)
By symmetry about the y-axis, $x_{\text{cm}} = 0$.
For a uniform semicircular wire of radius R, the centre of mass lies along the axis of symmetry at $y_{\text{cm}} = \frac{2R}{\pi}$.
Hence, the coordinates are $\left(0, \frac{2R}{\pi}\right)$.

Sol 12. Option (B)
For a hollow cone without base, element is a ring of circumference $2\pi r$ and slant length $dl = \frac{dy}{\cos\alpha}$ where $r = \frac{R}{h}y$:

$$dm = \sigma(2\pi r)dl \propto y\,dy$$

Taking apex at origin $y = 0$:

$$y_{\text{cm}} = \frac{\int_0^h y(y\,dy)}{\int_0^h y\,dy} = \frac{\int_0^h y^2 dy}{\int_0^h y\,dy} = \frac{h^3/3}{h^2/2} = \frac{2}{3}h \quad \text{(from apex)}$$

Sol 13. Option (A)
A quadrant of a disc has complete symmetry between x and y. Slicing along either axis integrates identically to a semicircular disc:

$$x_{\text{cm}} = \frac{4R}{3\pi}, \quad y_{\text{cm}} = \frac{4R}{3\pi}$$

Sol 14. Option (C)
The surface density $\sigma(r)$ depends only on the radial distance r, meaning the mass distribution possesses complete rotational symmetry about the origin.
Therefore, by symmetry, the centre of mass remains strictly at the geometric centre $(0, 0)$.

Sol 15. Option (B)
For the ring with $\lambda(\theta) = \lambda_0 \cos^2\theta$:

$$x_{\text{cm}} \propto \int_0^{2\pi} (R\cos\theta)(\lambda_0\cos^2\theta)R\,d\theta = \lambda_0 R^2 \int_0^{2\pi} \cos^3\theta\,d\theta = 0$$

$$y_{\text{cm}} \propto \int_0^{2\pi} (R\sin\theta)(\lambda_0\cos^2\theta)R\,d\theta = \lambda_0 R^2 \int_0^{2\pi} \sin\theta\cos^2\theta\,d\theta = 0$$

Thus, the centre of mass lies at $(0, 0)$.

Sol 16. Option (A)
Let the common base lie on the xy-plane at $z = 0$.
1. Solid hemisphere (in $-z$ region):
Volume $V_1 = \frac{2}{3}\pi R^3$, COM at $z_1 = -\frac{3}{8}R$.
2. Solid cone (in $+z$ region):
Volume $V_2 = \frac{1}{3}\pi R^2 h$, COM at $z_2 = +\frac{h}{4}$.
For COM of combined system to be at $z = 0$:

$$V_1 z_1 + V_2 z_2 = 0 \implies \left(\frac{2}{3}\pi R^3\right)\left(-\frac{3}{8}R\right) + \left(\frac{1}{3}\pi R^2 h\right)\left(\frac{h}{4}\right) = 0$$

$$\frac{h^2}{12} = \frac{6}{24}R^2 = \frac{1}{4}R^2 \implies h^2 = 3R^2 \implies \frac{h}{R} = \sqrt{3}$$

Sol 17. Option (A)
The centre of mass is a mathematical point and does not need to lie within the physical material of the body (e.g., for a ring, hollow sphere, hollow cone, or horseshoe, the COM lies in the empty hollow space). Hence statement (A) is false/incorrect.

Sol 18. Option (C)
Let the base of the cylinder be at $z = 0$.
1. Cylinder of length R ($0 \le z \le R$):
Volume $V_c = \pi R^2(R) = \pi R^3$, COM at $z_c = \frac{R}{2}$.
2. Hemispherical cap on top ($R \le z \le 2R$):
Volume $V_h = \frac{2}{3}\pi R^3$, COM at $z_h = R + \frac{3}{8}R = \frac{11}{8}R$.
Combined COM position:

$$z_{\text{cm}} = \frac{V_c z_c + V_h z_h}{V_c + V_h} = \frac{(\pi R^3)\left(\frac{R}{2}\right) + \left(\frac{2}{3}\pi R^3\right)\left(\frac{11}{8}R\right)}{\pi R^3 + \frac{2}{3}\pi R^3}$$

$$z_{\text{cm}} = \frac{\frac{1}{2} + \frac{11}{12}}{1 + \frac{2}{3}}R = \frac{\frac{17}{12}}{\frac{5}{3}}R = \frac{17}{12} \times \frac{3}{5}R = \frac{17}{20}R$$

Wait: $\frac{1}{2} + \frac{11}{12} = \frac{6+11}{12} = \frac{17}{12}$.
$\frac{17/12}{5/3} = \frac{17}{20}R$. Option (A) is $\frac{17}{20}R$. Let's ensure consistency: $z_{\text{cm}} = \frac{17}{20}R$.

Sol 19. Option (D)
Given $\lambda(x) = ax^2$:

$$x_{\text{cm}} = \frac{\int_0^L x(ax^2)dx}{\int_0^L ax^2 dx} = \frac{\int_0^L x^3 dx}{\int_0^L x^2 dx} = \frac{L^4/4}{L^3/3} = \frac{3}{4}L$$

Sol 20. Option (B)
Initial COM of the $2a \times 2b$ rectangle is at the geometric centre (a, b) relative to a corner $(0, 0)$.
When folded in half along the line $x = a$, the remaining half has length a in the x-direction with COM at $(a/2, b)$.
The magnitude of shift along x is:

$$\Delta x_{\text{cm}} = a - \frac{a}{2} = \frac{a}{2}$$